

Otherwise, `update()` calculates how much time has elapsed since the current image started showing by getting the current time using the system `time.time()` function and subtracting that from the time at which the current image started showing.

If the elapsed time is greater than the amount of time that each image should be showing, it's time to move to the next image. In this case, we increment `self.index` so that the upcoming call to the `draw()` method will draw the appropriate image. We then check whether the animation has finished. If not, we save the start time for the new image. If the animation is done, we set `self.playing` back to `False` (to indicate that we are no longer playing the animation), and we reset `self.index` to 0 so that the `draw()` method will show the first image again.

Finally, we call `draw()` in every frame ❸ to draw the current image of the animation. The `draw()` method assumes that `self.index` has been set correctly by a previous method, and uses it to index into the list of images. It then draws that image in the window at the specified location.

Example Main Program

[Listing 14-2](#) shows a main program that creates and uses a `SimpleAnimation` object. This will animate a dinosaur riding a bicycle.

File: `SimpleAnimation/Main_SimpleAnimation.py`

```
# Animation example
# Shows example of SimpleAnimation object

# 1 - Import library
import pygame
from pygame.locals import *
import sys
import pygamewidgets
from SimpleAnimation import *

# 2 Define constants
SCREEN_WIDTH = 640
SCREEN_HEIGHT = 480
FRAMES_PER_SECOND = 30
BGCOLOR = (0, 128, 128)
```